

CHINNASURYA PRASAD VULAVALA

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EXPERIENCE

- AI Software Engineer, Humanitarians AI, *Boston, MA*

June 2026 - Present

 - Built a **RAG** pipeline with a custom **Mixture-of-Experts** routing layer on top of **GPT-4o-mini**, routing student queries to specialist modes and injecting vector-searched textbook sources before inference.
 - Architected **LTI 1.3** and **Canvas** integration for an AI-powered education platform, enabling **OIDC-based SSO**, student provisioning, and grade synchronization via Assignment & Grade Services.
- Graduate Teaching Assistant, Northeastern University, *Boston, MA*

Sep 2025 - Apr 2026

 - Assisted 150+ students with **algorithm implementation**, **class design**, and **object-oriented programming** in **Java**, guiding hands-on lab sessions and improving submission quality by **30%**.
 - Mentored students and peer teaching assistants through 1:1 office hours, leading code reviews on **debugging**, **design patterns**, and refactoring techniques, reducing logical errors in final projects by **25%**.
- Software Engineer, UnitedHealth Group, *Hyderabad, India*

Aug 2023 - Jul 2024

 - Developed a **secure microservices** based messaging platform for care advisors using **Java** and **Spring Boot**, enabling seamless communication for 500K+ users, increasing system throughput by **50%**.
 - Implemented distributed messaging using **Apache Kafka**, building event-driven architecture for asynchronous member enrollment and downstream services, reducing API response time by **45%**.
 - Reworked search **APIs** by improving **PostgreSQL indexing** and **pagination** strategies for high-volume member data retrieval, reducing profile lookup time by **70%** and improving overall search responsiveness.
 - Optimized rendering through **React.memo**, **useMemo**, **useCallback**, and route-based code splitting with **React.lazy** and **Suspense**, reducing page load time by **45%** and improving Core Web Vitals by **35%**.
- Associate Software Engineer, UnitedHealth Group, *Hyderabad, India*

Aug 2022 - Jul 2023

 - Developed responsive user interfaces in **React** on the care advisor platform, building reusable components with **custom hooks** for data fetching and form handling, reducing duplicate code by **40%**.
 - Deployed services on **AWS** and **GCP**, provisioning infrastructure using **Terraform** with **auto-scaling** policies, reducing downtime, and optimizing resource allocation, cutting infrastructure costs by **15%**.
 - Designed automated deployment workflows with **GitHub Actions** and **Docker** for user-facing healthcare communication platforms, reducing manual deployment effort by 95% and enabling continuous delivery.
 - Built a security framework using **OAuth 2.0** and **JWT** for **REST APIs**, enforcing **role-based access control** across distributed backend services and reducing unauthorized access attempts by **70%**.

PROJECTS

- Distributed Key-Value Store | *Java, Paxos, gRPC, Docker, Kubernetes*

 - Engineered a **fault-tolerant distributed key-value** store using **Paxos**, ensuring strong consistency across replicated nodes and optimizing read/write operations by **30%** with efficient communication and caching.
- InterviewCoach AI | *Python, FastAPI, Vertex AI, Gemini Live API, React*

 - Designed an **LLM** based intelligent agent integrating external **Vertex AI** and **Gemini Live API** services for real time conversational feedback with prompt orchestration, guardrails, and evaluation metrics.

EDUCATION

- Master of Science in Computer Software Engineering

Aug 2024 - Apr 2026

Northeastern University, Boston, MA

GPA: 3.79 / 4.0
- Bachelor of Technology in Electronics and Telecommunication Engineering

Jul 2018 - May 2022

National Institute of Technology, Raipur

GPA: 8.65 / 10

TECHNICAL SKILLS

- Programming Languages:

C, CPP, Java, Python, Javascript, Typescript, SQL
- Software Development:

Spring Boot, Node.js, Express, HTML, CSS, React, Angular, Redux, REST, GraphQL
- AI & ML:

NLP, Transformers, RAG, LLMS, Fine-Tuning, Agentic AI, LangChain, LangGraph
- Databases & Messaging:

PostgreSQL, MySQL, MongoDB, ChromaDB, Redis, Kafka
- Cloud & Devops:

AWS, GCP, CI/CD, Github Actions, Docker, Terraform, Kubernetes